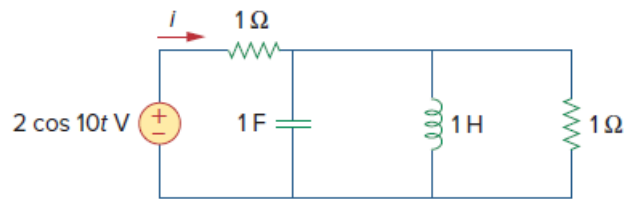


Problem

Determine i in the circuit of Fig. 10.50.

Figure 10.50:



Step-by-step solution

Step 1 of 3

Refer to the circuit diagram depicted in Figure 10.50 in the textbook.

The value of voltage source is,

$$v_s(t) = 2 \cos 10t \text{ V}$$

Represent the voltage source in phasor domain.

$$\mathbf{V}_s = 2 \angle 0^\circ \text{ V}$$

The value of angular frequency is, $\omega = 10 \text{ rad/s}$

Calculate the inductive reactance of 1 H inductor.

$$\begin{aligned} X_L &= j\omega L \\ &= j(10)(1) \\ &= j10 \Omega \end{aligned}$$

Calculate the capacitive reactance of 1 F capacitor.

$$\begin{aligned} X_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(10)(1)} \\ &= -j0.1 \Omega \end{aligned}$$

Step 2 of 3

Draw the circuit diagram in frequency domain.

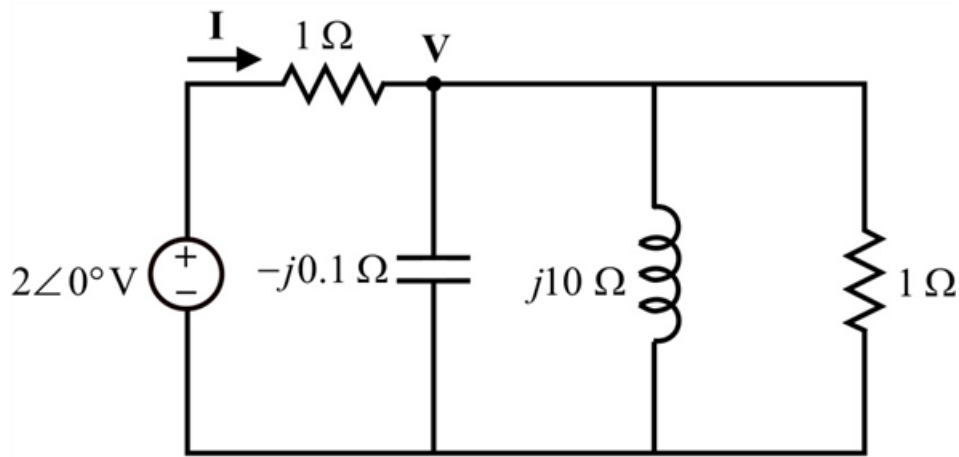


Figure 1

Step 3 of 3

Apply Kirchhoff's current law at node $\mathbf{V}$.

$$\frac{\mathbf{V} - 2\angle 0^\circ}{1} + \frac{\mathbf{V}}{-j0.1} + \frac{\mathbf{V}}{j10} + \frac{\mathbf{V}}{1} = 0$$

$$\mathbf{V} \left(1 + \frac{1}{-j0.1} + \frac{1}{j10} + 1 \right) = 2\angle 0^\circ$$

$$\mathbf{V} (10.1\angle 78.58^\circ) = 2\angle 0^\circ$$

$$\mathbf{V} = 0.198\angle -78.58^\circ \text{ V}$$

From Figure 1, determine the value of current, $\mathbf{I}$.

$$\begin{aligned} \mathbf{I} &= \frac{2\angle 0^\circ - \mathbf{V}}{1} \\ &= 2\angle 0^\circ - 0.198\angle -78.58^\circ \\ &= 1.9704\angle 5.65^\circ \text{ A} \end{aligned}$$

Convert the value of current $\mathbf{I}$ from phasor domain to time domain.

$$i(t) = 1.9704 \cos(10t + 5.65^\circ) \text{ A}$$

Therefore, the value of current, i in the circuit is $1.9704 \cos(10t + 5.65^\circ) \text{ A}$.